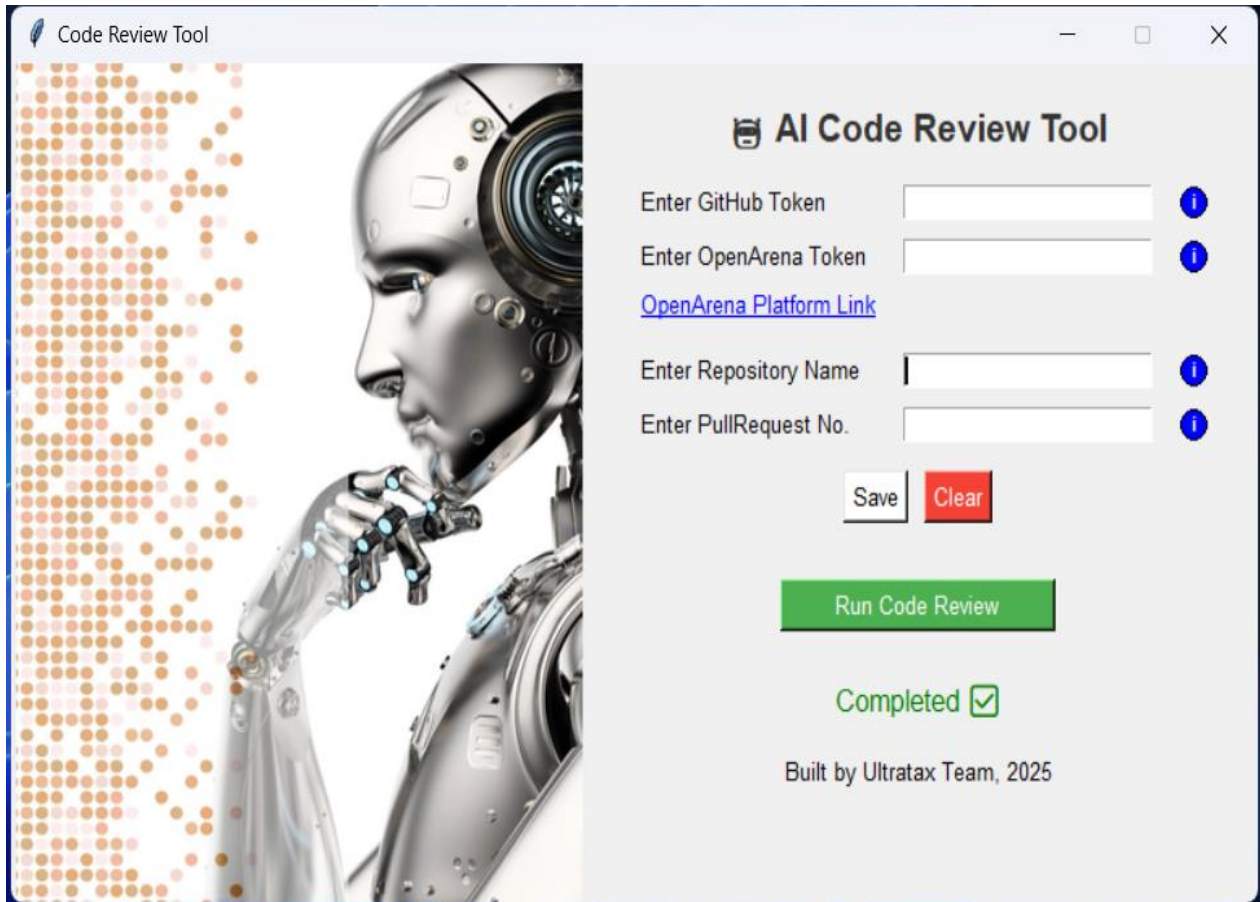


AI Code Review Tool

This Python-based tool leverages the GitHub API and OpenArena AI to automate code reviews on pull requests. Designed exclusively for Thomson Reuters employees, it enhances the code review process by extracting modified lines, sending them for AI-based analysis, and posting insightful comments and suggestions directly on GitHub PRs.



The screenshot shows a web application window titled "Code Review Tool". On the left is a large image of a silver robot head in profile, with a background of orange and white dots. The right side of the window contains the "AI Code Review Tool" interface. It has five input fields: "Enter GitHub Token", "Enter OpenArena Token", "Enter Repository Name", and "Enter PullRequest No.", each with an information icon (i) to its right. There is also a blue link labeled "OpenArena Platform Link". Below the inputs are "Save" and "Clear" buttons. A large green button labeled "Run Code Review" is positioned below the "Save" and "Clear" buttons. At the bottom, the text "Completed" is displayed in green with a green checkmark icon. The footer text reads "Built by Ultratax Team, 2025".

Code Review Tool

AI Code Review Tool

Enter GitHub Token

Enter OpenArena Token

[OpenArena Platform Link](#)

Enter Repository Name

Enter PullRequest No.

Save Clear

Run Code Review

Completed ☒

Built by Ultratax Team, 2025

🔧 Installation

1 Prerequisites

- Ensure you have the following installed:

Supports for Windows only

Download Python:

- Visit the [official Python website](https://www.python.org/) and download the latest version of Python 3.8+. Run the installer. Ensure you check the box that says "Add Python to PATH" during installation.

Install pip:

- Pip is automatically installed with Python 3.8+. You can verify the installation by opening Command Prompt and typing:

```
python --version  
pip --version
```

- If pip is not installed, you can manually install it by downloading the get-pip.py script from [here](https://bootstrap.pypa.io/get-pip.py) and running:

```
python get-pip.py
```

⚠ NOTE : Install pyinstaller and run this if any changes made in the code logic:

```
pip install pyinstaller  
pyinstaller --onefile .\AIReview.py
```

- Install PyGithub

```
pip install PyGithub requests
```

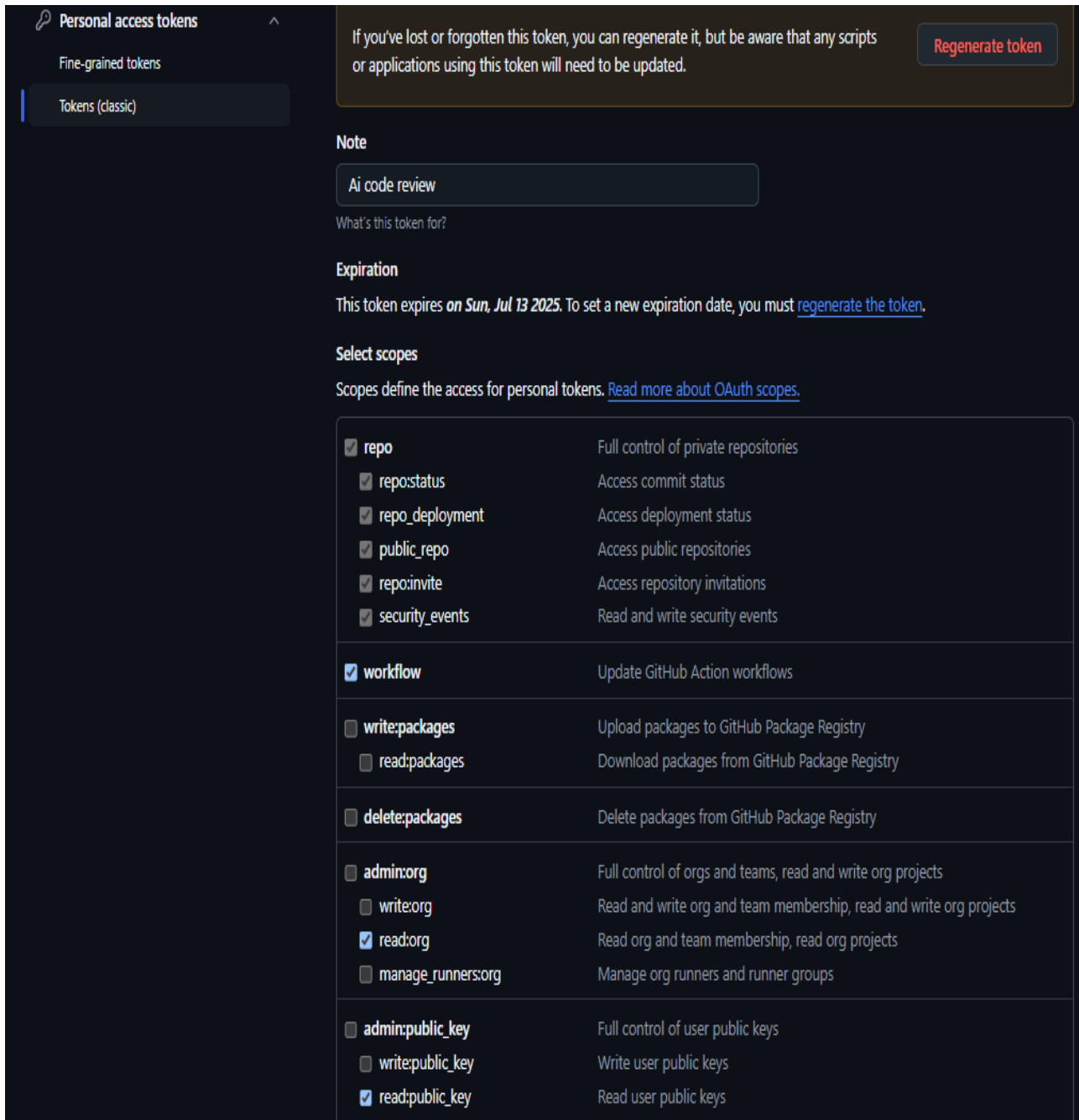
2 Get the Open Arena token

- Please refer 🖱️ [Open Arena Link](#)

3 Create Github Token

Generate a GitHub Token:

- Navigate to the developer settings on GitHub and create a token. Please choose the "Classic token" option.
- Ensure the Following Options are Selected:



The screenshot shows the GitHub 'Personal access tokens' settings page. On the left, a sidebar contains 'Personal access tokens', 'Fine-grained tokens', and 'Tokens (classic)'. The main content area has a dark theme. At the top, a warning box states: 'If you've lost or forgotten this token, you can regenerate it, but be aware that any scripts or applications using this token will need to be updated.' with a 'Regenerate token' button. Below this is a 'Note' section with a text input containing 'Ai code review' and a label 'What's this token for?'. The 'Expiration' section shows 'This token expires on Sun, Jul 13 2025. To set a new expiration date, you must [regenerate the token](#).' The 'Select scopes' section includes a link 'Read more about OAuth scopes.' and a list of scopes with checkboxes. The selected scopes are: repo, repo:status, repo_deployment, public_repo, repo:invite, security_events, workflow, read:org, and read:public_key.

Personal access tokens

Fine-grained tokens

Tokens (classic)

If you've lost or forgotten this token, you can regenerate it, but be aware that any scripts or applications using this token will need to be updated. [Regenerate token](#)

Note

What's this token for?

Expiration

This token expires **on Sun, Jul 13 2025**. To set a new expiration date, you must [regenerate the token](#).

Select scopes

Scopes define the access for personal tokens. [Read more about OAuth scopes.](#)

☒ repo	Full control of private repositories
☒ repo:status	Access commit status
☒ repo_deployment	Access deployment status
☒ public_repo	Access public repositories
☒ repo:invite	Access repository invitations
☒ security_events	Read and write security events
☒ workflow	Update GitHub Action workflows
☐ write:packages	Upload packages to GitHub Package Registry
☐ read:packages	Download packages from GitHub Package Registry
☐ delete:packages	Delete packages from GitHub Package Registry
☐ admin:org	Full control of orgs and teams, read and write org projects
☐ write:org	Read and write org and team membership, read and write org projects
☒ read:org	Read org and team membership, read org projects
☐ manage_runners:org	Manage org runners and runner groups
☐ admin:public_key	Full control of user public keys
☐ write:public_key	Write user public keys
☒ read:public_key	Read user public keys

☐ admin:repo_hook	Full control of repository hooks
☐ write:repo_hook	Write repository hooks
☐ read:repo_hook	Read repository hooks
☒ admin:org_hook	Full control of organization hooks
☐ gist	Create gists
☒ notifications	Access notifications
☐ user	Update ALL user data
☒ read:user	Read ALL user profile data
☒ user:email	Access user email addresses (read-only)
☐ user:follow	Follow and unfollow users
☐ delete_repo	Delete repositories
☐ write:discussion	Read and write team discussions
☒ read:discussion	Read team discussions
☐ admin:enterprise	Full control of enterprises
☐ manage_runners:enterprise	Manage enterprise runners and runner groups
☐ manage_billing:enterprise	Read and write enterprise billing data
☐ read:enterprise	Read enterprise profile data
☐ scim:enterprise	Provisioning of users and groups via SCIM
☐ audit_log	Full control of audit log
☐ read:audit_log	Read access of audit log
☐ codespace	Full control of codespaces
☐ codespace:secrets	Ability to create, read, update, and delete codespace secrets
☐ copilot	Full control of GitHub Copilot settings and seat assignments
☐ manage_billing:copilot	View and edit Copilot Business seat assignments
☐ write:network_configurations	Write org hosted compute network configurations
☐ read:network_configurations	Read org hosted compute network configurations
☐ project	Full control of projects
☒ read:project	Read access of projects

☐ admin:gpg_key	Full control of public user GPG keys
☐ write:gpg_key	Write public user GPG keys
☒ read:gpg_key	Read public user GPG keys
☐ admin:ssh_signing_key	Full control of public user SSH signing keys
☐ write:ssh_signing_key	Write public user SSH signing keys
☐ read:ssh_signing_key	Read public user SSH signing keys

Update token [Cancel](#)

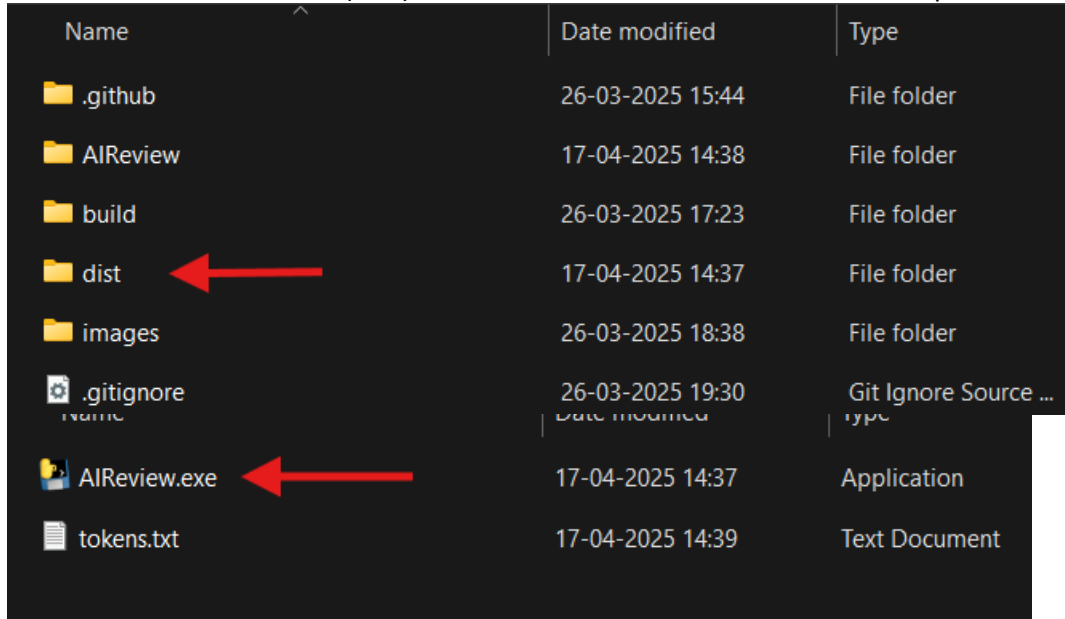
Configuration

- ! Before running the tool, make sure you have:
- GitHub Token: For authentication with the GitHub API
 - OpenArena Token: To send modified code for AI-based review
 - These credentials must be entered when prompted in the GUI.
-

Usage

1. Running the Tool:

- You can find the executable (.exe) for the tool in the **dist** folder. See the examples below:



Name	Date modified	Type
.github	26-03-2025 15:44	File folder
AIReview	17-04-2025 14:38	File folder
build	26-03-2025 17:23	File folder
dist	17-04-2025 14:37	File folder
images	26-03-2025 18:38	File folder
.gitignore	26-03-2025 19:30	Git Ignore Source ...
AIReview.exe	17-04-2025 14:37	Application
tokens.txt	17-04-2025 14:39	Text Document

2. Enter the required details in the GUI:

- GitHub Token
- OpenArena Token
- Repository Name (e.g., username/repo)
- Pull Request Number

3. Click "Run Code Review" to initiate the process.

4. AI-generated comments will be posted directly on the PR.

5. Check your PR on GitHub to view the feedback.

Features

-  Extracts exact modified lines from PR patches
-  Sends changes to OpenArena AI for review
-  Analyzes logic impact, potential issues, and code consistency
-  Posts AI-generated comments on GitHub PR
-  Displays progress and results in a simple Tkinter GUI

🔧 Troubleshooting

- ♦ Error: Authentication failed – Ensure your GitHub token has the correct permissions.
- ♦ Error: AI review failed – Check your OpenArena API token and internet connection.
- ♦ Comments not appearing on PR? – Verify that the PR number and repository name are correct.

For any issues, feel free to open an issue in the repo.

🏆 Credits

Developed by the Ultratax Team, 2025.

- **Kalyani, Kandunuri**
 - **Harish Sarma, Velavalapalli**
 - Special thanks to **Radhika Ramagiri** and **Prasad Kolaparthi** ❤️
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📚 Resources Used

- GitHub API Documentation: For understanding how to interact with GitHub programmatically.
- OpenArena API Documentation: For integrating AI-based code analysis.
- Python Official Documentation: For language-specific features and libraries.
- Tkinter Documentation: For creating the GUI interface.
- PyGitHub Documentation: For utilizing GitHub API features.